

ALTIRIG · REQUIREMENTS

AltiRig requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-03	Amish Chadha	Amish's decisions 41B (80 % open baffle; R6 met on paper at 1.8 % to 3.3 %) and 42B (surplus pipe and heads, host lab's DC supply and pump; R10 restated to USD 4,000 and met at USD 3,865), ALR-DDR-003
0.3	2026-10-03	Amish Chadha	TRL 3 status from the calculations (ALR-CAL-001) and the constructable design (ALR-DDR-002)
0.2	2026-10-03	Amish Chadha	TRL 2 review; concept status for every requirement; R11 interlocks and R12 over-vacuum protection added (ALR-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

AltiRig requirements

All twelve requirements are met on paper by the constructable design, R10 as restated on 2026-10-03. Targets are unchanged from the scaffold except for the two safety requirements added at the TRL 2 review (R11 and R12) and R10, restated by Amish's decision 42B. Status is on paper only: every requirement is verified by test at TRL 4.

On 2026-10-03 Amish decided the two requirement decisions put to him at TRL 3 as recommended. Amish, 2026-10-03: "i agree with all the 46 recommendations you provided. please proceed." For AltiRig these are 41B (R6: an 80 % open baffle) and 42B (R10: a surplus pipe and heads and the host lab's DC supply and pump, with the cost target restated), recorded in ALR-DDR-003.

Table 1. Requirements and their status at TRL 3.

ID	Requirement	Target	Verification (TRL 4 or later)	Status at TRL 3
R1	Density range	Hold any air density from ambient down to 0.74 kg/m ³ (5,000 m equivalent) at room temperature	Pressure and temperature log over a 10 min hold at several set points	Met on paper: 0.736 kg/m ³ needs 61.9 kPa at 20 °C; the pump reaches far lower and the relief valve stops at 46.3 kPa (ALR-CAL-001, A)
R2	Density control	Density held within plus or minus 1% of set point during a run	Logged density during motor runs	Met on paper: error budget 0.6 % (ALR-CAL-001, D)
R3	Thrust range and accuracy	0 to 50 N thrust with error no more than 1% of full scale	Calibration with certified masses before and after a test series	Met on paper: flexure stand with a 10 kg cell, error budget 0.13 N, 0.27 % (ALR-CAL-001, E)
R4	Torque and speed	Torque 0 to 2 N m within 2%; speed within 0.5%	Calibration arm with masses; optical tachometer cross-check	Met on paper: torque error 0.3 %, speed 0.02 % (ALR-CAL-001, E)
R5	Propeller size	Propellers up to 380 mm (15 in) diameter	Fit check and clearance measurement	Met on paper: 210 mm from tip to wall, 80 mm to the deck (ALR-CAL-001, E)
R6	Chamber effect	At ambient pressure, chamber thrust within 5% of an open-air stand for the same propeller and speed	Side-by-side comparison tests	Met on paper (decision 41B): estimated 1.8 % to 3.3 % high with the 80 % open baffle, was 2.9 % to 5.3 % with the 63 % baffle (ALR-CAL-001, F); confirmed by the TRL 4 comparison runs with and without the baffle
R7	Vessel safety factor	Collapse pressure at least 4 times the maximum working pressure difference (target)	Calculation reviewed by a qualified engineer, then proof test behind a guard	Met on paper, at full vacuum: shell 6.7, heads 40, window 7.1 on yield (ALR-CAL-001, B)
R8	Pump-down time	Ambient to 54 kPa in no more than 5 min	Timed pump-down with pressure log	Met on paper: 4.0 min with a 170 L/min pump (ALR-CAL-001, C)
R9	Data output	Results saved as CSV with density, thrust, torque, speed, voltage and current at 10 Hz or more	Inspection of logged files	Met on paper: 20 Hz logging, amplifiers at 80 Hz (ALR-CAL-001, G)
R10	Cost	Restated 2026-10-03 (decision 42B): complete rig at or below USD 4,000, built with a surplus pipe offcut and tank heads that pass the roundness and thickness checks, and the host lab's existing DC supply and vacuum pump; the USD 1,000 value-	Costed bill of materials	Met as restated: USD 3,865, USD 135 under USD 4,000 and USD 2,865 over the value-engineering target (ALR-CAL-001, H)

ID	Requirement	Target	Verification (TRL 4 or later)	Status at TRL 3
		engineering target is kept as a control figure. Was: complete rig at or below USD 1,000		
R11	Interlocks	The motor can be powered only with the door latched and the arming key turned; the emergency stop removes motor and pump power; the bleed valve opens and vents the chamber on loss of power	Function check of each interlock before first power	Met on paper by design (ALR-BLD-001, sections 4 and 6); the door is also held shut by 24 kN of air load at the 5,000 m point
R12	Over-vacuum protection	A relief valve stops the chamber going below 46.3 kPa absolute (55 kPa below atmosphere); the vessel is still designed for full vacuum	Relief valve set and checked against the gauge before first pump-down	Met on paper (ALR-CAL-001, A and B)

■ Requirements not met or at risk

None on paper. The two that were (R6 at risk, R10 not met) were decided by Amish on 2026-10-03: Amish, 2026-10-03: "i agree with all the 46 recommendations you provided. please proceed."

- **R6, decision 41B.** The closed loop of air inside the vessel adds a little resistance to the propeller's own flow, so the propeller is expected to give slightly more thrust in the chamber than in open air. With the 80 % open baffle two simple models give 1.8 % and 3.3 %, inside the 5 % target. The TRL 4 comparison runs decide it; the 1,016 mm vessel is held in reserve if they show more than 5 %.
- **R10, decision 42B.** Restated to USD 4,000 with surplus vessel steel and the host lab's DC supply and pump; estimated USD 3,865. It depends on the surplus pipe and heads passing the roundness and thickness checks, and on the host lab having a DC supply of at least 1,500 W and a pump of at least about 140 L/min. Buying a new pipe, heads, pump and supply instead would cost about USD 4,930; buying only a new pump and supply, about USD 4,400.

■ Assumptions

- Matching air density is the main effect; the small differences in viscosity and speed of sound between a room-temperature chamber and real air at 5,000 m and -17.5 °C are second order for small propellers (to be checked). Blade Reynolds number falls by about 39 % at 5,000 m density (ALR-CAL-001, E), which the chamber reproduces because density, not pressure, is matched.
- A surplus or new-cut 813 mm steel pipe offcut and two tank heads can be bought; the out-of-roundness of the pipe is no more than 1 % of its diameter (to confirm when bought).
- The chamber effect is measured by side-by-side runs at ambient pressure and recorded with every published data set.
- Earlier physics estimate: hover time at 5,000 m and -20 °C is about 47 % of sea level; AltiRig will test the aerodynamic share of that estimate.

Safety: R7, R11 and R12 are safety requirements. They are met on paper only; no pump-down or motor run is made until the safety stops in the build plan (ALR-BLD-001, section 6) are passed and a qualified engineer has reviewed the vessel calculation.